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Brief Communication

SEASONAL VARIABILITY IN BEHAVIORAL DESPAIR IN FEMALE RATS

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Separate groups of female Wistar rats were tested every three months over a year to assess the extent of seasonal variability in the behavioral despair model of depression consisting of 2 swim tests separated by 24 h. There was a significant seasonal variability in the duration of immobility in the swim tests; duration of immobility was shorter in August and November than in February and May. These findings suggest reduced immobility and susceptibility to behavioral despair in summer and fall compared to winter and spring.

Keywords behavioral despair, forced swimming, seasonal depression

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A large number of studies have reported seasonal variability in mood disorders, depression, and related neurochemical mechanisms (Abel, 1995; Amat & Torres, 1993; Arora et al., 1984; Bissette et al., 1995; Carlsson et al., 1980; Kasper et al., 1989; Pande et al., 1992; Parker & Walter, 1982; Weiner et al., 1992). In particular, one form of depression, seasonal affective disorder (SAD), is marked by strong seasonal variability. In SAD, episodes of depression regularly occur in winter and fall with symptom remission in spring and summer (Lewy et al., 1982; Rosenthal et al., 1984). These findings point to the importance of understanding the mechanisms underlying seasonal variations in psychopathology to be able to assess the relative contributions of environmental and endogenous factors (Mrosovsky; 1988; Young et al., 1997). An approach to assess the contribution of endogenous factors to seasonal variations is to study animals kept in constant environmental conditions. A survey of the literature indicates a paucity of animal studies on seasonal variability in animal models of depression. One example is the investigation of seasonal variability in immobility in forced swimming that indicated that rats display significantly longer immobility in late fall and winter than in spring and summer (Abel, 1995). This finding is potentially important for behavioral despair, a frequently used animal model of depression based on forced swim tests (Porsolt et al., 1977, 1978; Willner, 1990). In the aforementioned study (Abel, 1995), animals were exposed to a single 11-min swim test. The present study investigated seasonal variation in immobility and other behaviors displayed in the more typical format of the behavioral despair model of depression. The model, originally developed by Porsolt (1977, 1978), is based on assessing the animal's immobility in two swim

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tests separated by 24 h: rats display longer immobility in the second swim test compared to the first.

MATERIALS AND METHODS

Subjects

A total of 70 female Wistar rats raised in the authors' breeding colony, weighing 195 to 245 g, were used in the experiments. Animals were group housed in plastic cages with free access to food and water. Subjects were maintained on a 12 h light:12 h dark schedule of lighting (lights on at 7 AM) in the colony room kept at 20 ± 2ºC.

Procedure

Groups of rats (n = 4 each) were run in forced swim tests in the first 10 days of May, August, November, and February. The results of the first group were replicated with a group of 14 female rats a year later (the May 2 group). All testing was done between 11:00 and 15:00. Subjects in all groups were tested in 2 forced swim tests; a 15-min test followed 24 h later by a 5-min test. For the tests, the subject was placed in a Plexiglas cylinder (45 cm height, 30 cm diameter) filled with 25°C water to a height of 15 cm. After the test, the subject was placed under a lamp for 30 min for drying and was then returned to its home cage. Swim tests were recorded on videotape.

Statistical Analysis

In the second swim test, total duration of immobility and the first time the subject became immobile for 5 s or longer (denoted as "onset of immobility") were analyzed. Immobility was defined as floating or remaining motionless without leaning against the wall of the cylinder. One-way analysis of variance was used to evaluate monthly differences in duration and onset of immobility. Significant differences ($p < .05$) were determined by post hoc comparisons based on Newman-Keuls tests.

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RESULTS

Duration of Immobility

Mean duration of immobility in the second swim test is shown in Figure 1 for all the groups. There was a significant effect of month of testing [$F(4, 65) = 4.516$, $p < .01$]. Animals tested in August and November displayed significantly less immobility compared to the other three groups.

Onset of Immobility

The first time subjects became immobile for at least 5 consecutive seconds is shown in Figure 2. The monthly differences in onset of immobility failed to reach significance.

DISCUSSION

The present study indicates a seasonal variability in the behavior of female Wistar rats in forced swim tests. Overall, animals tested in August and November showed less immobility than those tested in

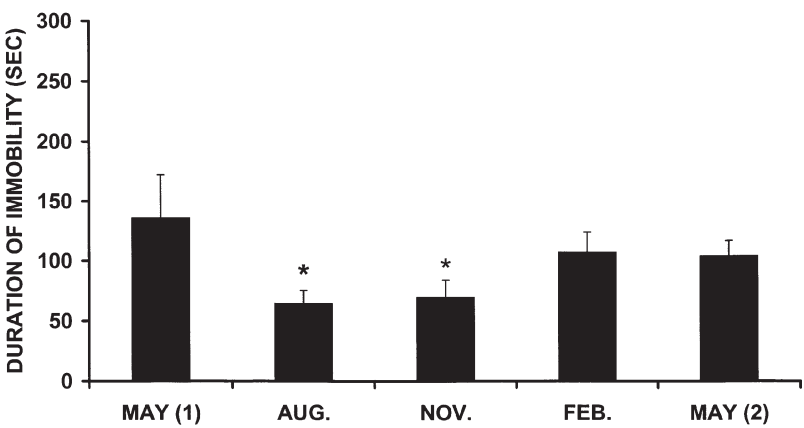


FIGURE 1. Duration of immobility (mean ± SEM) in the second of two forced swim tests during different months of the year. Asterisk indicates a significant difference ($p < .05$) with the months that are not marked with an asterisk.

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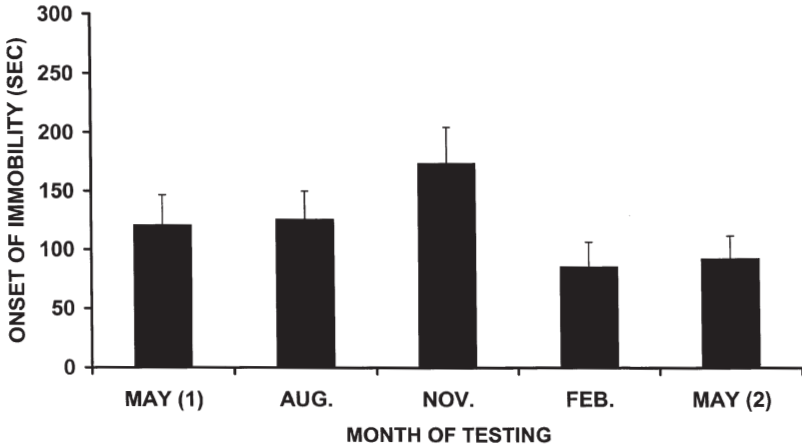


FIGURE 2. Onset of immobility (mean ± SEM)—the first time subjects became immobile for at least 5 consecutive s—in the second forced swim test during different months of the year.

February and May. These findings therefore indicate that animals tested in the summer and fall months seem to be less susceptible to behavioral despair than those tested in winter and spring. The present results corroborate and extend the findings that male Sprague-Dawley rats subjected to a single 11-min forced swim test displayed longer immobility in the winter than in the summer months (Abel, 1995).

An additional behavioral measure to assess behavioral despair in the present study was onset of immobility defined as latency to become immobile for at least 5 consecutive s. Although failing to reach significance, monthly differences in this behavioral measure also showed a pattern that paralleled the monthly differences in total duration of immobility: onset of immobility occurred later in the second swim test in August and November than in February and May. This finding strengthens the argument that behavioral despair may be attenuated in summer and fall.

Because environmental conditions in the laboratory were kept constant throughout the year, the source of seasonal variability in the immobility scores has to be sought elsewhere. It is possible, for example, that an endogenously generated circannual rhythm in locomotor activity may account for the observed seasonal differences in immobility scores in the present study. Although this is a

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possibility, it is suggestive that male Wistar rats do not display a circannual locomotor rhythm (Nagayama & Lu, 1998).

Reported seasonal differences in the production of neurotransmitters and their receptors may account for the neurochemical basis of the observed monthly differences in behavioral despair observed in the present study (Bissette et al., 1995; Carlsson et al., 1980, Connor et al., 1997; Nagayama & Lu, 1998; Weiner et al., 1992). Another possible explanation of the present findings may be based on the putative involvement of the suprachiasmatic nuclei (SCN) in the observed behavioral variability. SCN are involved in circannual regulation of behavioral and biological rhythms (Rusak & Zucker, 1979; Turek, 1985) and may play a role in depression (Van Cauter & Turek, 1986). Studies also point to seasonal structural variability in SCN in both humans (Hofman & Swaab, 1992) and rats (Peschke et al., 1996). Neuroanatomical studies have revealed extensive and usually reciprocal relationship between the SCN and several limbic structures that figure critically in affective behavior and psychopathology (Buijs et al., 1998; Leak & Moore, 2001; Vrang et al., 1995; Watts & Swanson, 1987; Watts et al., 1987). Moreover, the present authors have recently demonstrated that bilateral destruction of the SCN in male Wistar rats has a protective effect in behavioral despair (Tataroglu et al., 2004).

In conclusion, these results suggest that there is significant seasonal variability in immobility in the behavioral despair test. These findings parallel observations in humans that some forms of depression show seasonal variability (Kasper et al., 1989; Pande et al., 1992; Parker & Walter, 1982; Rosenthal et al., 1984). The present findings, therefore, not only provide a behavioral basis for the investigation of seasonal variability in mechanisms underlying behavioral despair, but should also be interpreted as a cautionary note in using immobility scores in assessing results of experiments conducted at different times of the year.

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